

Integration — Lecture Notes

Calculus · BSCS Semester 3 · Week 5, Lectures 1–2 — Definite/Indefinite Integrals, Double Integrals, Triple Integrals

1. What Is Integration?

Integration is the process of finding the **total accumulation** of a quantity, or the **original function**, when only its **rate of change** is known. In this sense it is the reverse (inverse) operation of differentiation.

Integration is commonly used to find:

- Distance covered, given a velocity (rate of change of position)
- Area of a region
- The original function, given its derivative (inverse of derivative)
- The area under a curve, between the curve and the x-axis

Derivatives vs. Integration

Derivatives	Integration
Expand a function — break it down into its instantaneous rate of change.	Accumulates / sums a function. The symbol $\int$ means “sum” or “accumulate.”

2. Definite vs. Indefinite Integration

Definite Integration	Indefinite Integration
• Has limits (an upper limit and a lower limit) • Starting and ending points are known • Evaluating it gives a single number (no “+ C”)	• Has no limits • The answer always contains a constant, “+ C”

Why “+ C”? Because when we integrate, we don’t know which constant existed in the original function before it was differentiated (a constant’s derivative is always 0). Adding “+ C” accounts for every possibility.

Definite Integral — General Form

$\int f(x) \, dx$, evaluated from a lower limit to an upper limit

Worked Example

$$\int_0^5 x \, dx = [x^2/2]_0^5 = 5^2/2 - 0^2/2 = 25/2$$

3. Double Integration

A double integral extends integration to a function of **two variables**, $f(x, y)$, integrated successively over both axes. It is written:

$$\iint f(x, y) \, dx \, dy$$

To evaluate it, integrate with respect to the **inner variable first** (treating the other variable as a constant), then integrate the result with respect to the outer variable.

Worked Example 1

$$\int_0^1 \int_0^2 (x + y) \, dx \, dy$$

Step 1 — Integrate w.r.t. x (inner integral, y held constant, limits $0 \rightarrow 2$):

$$\int (x + y) \, dx = x^2/2 + xy$$

$$\text{Apply limits: } [2^2/2 + 2y] - [0^2/2 + 0] = 2 + 2y$$

Step 2 — Integrate the result w.r.t. y (outer integral, limits $0 \rightarrow 1$):

$$\int (2 + 2y) \, dy = 2y + y^2$$

Step 3 — Apply the y -limits (0 to 1):

$$= [2(1) + 1^2] - [2(0) + 0^2]$$

$$= 2 + 1$$

Answer = 3

Worked Example 2

$$\int_0^1 \int_0^2 (x^2 + y^2) \, dx \, dy$$

Step 1 — Integrate w.r.t. x (inner integral, y held constant, limits $0 \rightarrow 2$):

$$\int (x^2 + y^2) \, dx = x^3/3 + xy^2$$

$$\text{Apply limits: } [2^3/3 + 2y^2] - [0] = 8/3 + 2y^2$$

Step 2 — Integrate the result w.r.t. y (outer integral, limits $0 \rightarrow 1$):

$$\int (8/3 + 2y^2) \, dy = 8y/3 + 2y^3/3$$

Step 3 — Apply the y -limits (0 to 1):

$$= [8(1)/3 + 2(1)^3/3] - [0]$$

$$= 8/3 + 2/3$$

Answer = $10/3$

Note: A definite (double) integral always gives a constant numeric answer; an indefinite one gives a functional (variable) answer.

Practice Problems — Double Integrals (Detailed Solutions)

$$\text{Q1. } \int_0^4 \int_0^3 (xy + x) \, dx \, dy$$

Step 1 — Integrate w.r.t. x , treating y as a constant (limits $0 \rightarrow 3$):

$$\int (xy + x) \, dx = x^2y/2 + x^2/2$$

$$\text{Apply limits: } [3^2y/2 + 3^2/2] - [0] = 9y/2 + 9/2$$

Step 2 — Integrate the result w.r.t. y (limits $0 \rightarrow 4$):

$$\int (9y/2 + 9/2) dy = 9y^2/4 + 9y/2$$

Step 3 — Apply the y-limits (0 to 4):

$$= [9(4)^2/4 + 9(4)/2] - [9(0)^2/4 + 9(0)/2]$$

$$= [9(16)/4 + 36/2] - 0$$

$$= 36 + 18$$

Answer = 54

Q2. $\int_0^1 \int_0^2 (x + y) dy dx$

Step 1 — Integrate w.r.t. y, treating x as a constant (limits $0 \rightarrow 2$):

$$\int (x + y) dy = xy + y^2/2$$

$$\text{Apply limits: } [x(2) + 2^2/2] - [0] = 2x + 2$$

Step 2 — Integrate the result w.r.t. x (limits $0 \rightarrow 1$):

$$\int (2x + 2) dx = x^2 + 2x$$

Step 3 — Apply the x-limits (0 to 1):

$$= [1^2 + 2(1)] - [0^2 + 2(0)]$$

$$= 1 + 2$$

Answer = 3

Q3. $\int_0^1 \int_0^2 (x^2 + y^2) dy dx$

Step 1 — Integrate w.r.t. y, treating x as a constant (limits $0 \rightarrow 2$):

$$\int (x^2 + y^2) dy = x^2y + y^3/3$$

$$\text{Apply limits: } [x^2(2) + 2^3/3] - [0] = 2x^2 + 8/3$$

Step 2 — Integrate the result w.r.t. x (limits $0 \rightarrow 1$):

$$\int (2x^2 + 8/3) dx = 2x^3/3 + 8x/3$$

Step 3 — Apply the x-limits (0 to 1):

$$= [2(1)^3/3 + 8(1)/3] - [0]$$

$$= 2/3 + 8/3$$

Answer = 10/3

4. Triple Integration

A triple integral extends the same idea to a function of **three variables**, $f(x, y, z)$, integrated successively over three axes (x , y , and z):

$$\iiint f(x, y, z) \, dx \, dy \, dz$$

As with double integrals, work from the **innermost** integral outward, integrating with respect to one variable at a time while treating the others as constants, and applying each pair of limits as you go.

Worked Example

$$\int_0^3 \int_0^3 \int_0^2 (x + y + z) \, dx \, dy \, dz$$

Step 1 — Integrate w.r.t. x (innermost, y and z held constant, limits $0 \rightarrow 2$):

$$\int (x + y + z) \, dx = x^2/2 + xy + xz$$

$$\text{Apply limits: } [2^2/2 + 2y + 2z] - [0] = 2 + 2y + 2z$$

Step 2 — Integrate the result w.r.t. y (middle integral, z held constant, limits $0 \rightarrow 3$):

$$\int (2 + 2y + 2z) \, dy = 2y + y^2 + 2zy$$

$$\text{Apply limits: } [2(3) + 3^2 + 2z(3)] - [0] = 6 + 9 + 6z = 15 + 6z$$

Step 3 — Integrate the result w.r.t. z (outer integral, limits $0 \rightarrow 3$):

$$\int (15 + 6z) \, dz = 15z + 3z^2$$

Step 4 — Apply the z -limits (0 to 3):

$$= [15(3) + 3(3)^2] - [0]$$

$$= 45 + 27$$

Answer = 72

Setting up limits: in a triple integral, each variable is assigned its own pair of limits (lower $\rightarrow$ upper). The innermost integral is evaluated first, its result is substituted into the next integral, and so on until a single number remains.

Practice Problems — Triple Integrals (Detailed Solutions)

Q1. $\int_0^1 \int_0^1 \int_0^1 x \, dx \, dy \, dz$

Step 1 — Integrate w.r.t. x (limits $0 \rightarrow 1$): $\int x \, dx = x^2/2$

$$\text{Apply limits: } [1^2/2] - [0] = 1/2$$

Step 2 — Integrate the constant $1/2$ w.r.t. y (limits $0 \rightarrow 1$):

$$\int (1/2) \, dy = y/2 \rightarrow [1/2] - [0] = 1/2$$

Step 3 — Integrate the constant $1/2$ w.r.t. z (limits $0 \rightarrow 1$):

$$\int (1/2) \, dz = z/2 \rightarrow [1/2] - [0] = 1/2$$

Answer = $1/2$

Q2. $\int_1^2 \int_1^2 \int_1^2 y \, dx \, dy \, dz$

Step 1 — Integrate w.r.t. x (y is constant w.r.t. x , limits $1 \rightarrow 2$):

$$\int y \, dx = xy \rightarrow [2y] - [1y] = y$$

Step 2 — Integrate y w.r.t. y (limits $1 \rightarrow 2$): $\int y \, dy = y^2/2$

$$\text{Apply limits: } [2^2/2] - [1^2/2] = 2 - 1/2 = 3/2$$

Step 3 — Integrate the constant 3/2 w.r.t. z (limits 1 → 2):

$$\int (3/2) dz = 3z/2 \rightarrow [3(2)/2] - [3(1)/2] = 3 - 3/2 = 3/2$$

Answer = 3/2

Q3. $\int_0^2 \int_0^3 \int_1^2 xy \, dx \, dy \, dz$

Step 1 — Integrate w.r.t. x (y constant, limits 1 → 2):

$$\int xy \, dx = x^2 y / 2$$

$$\text{Apply limits: } [2^2 y / 2] - [1^2 y / 2] = 2y - y/2 = 3y/2$$

Step 2 — Integrate 3y/2 w.r.t. y (limits 0 → 3):

$$\int (3y/2) \, dy = 3y^2 / 4$$

$$\text{Apply limits: } [3(3)^2 / 4] - [0] = 27/4$$

Step 3 — Integrate the constant 27/4 w.r.t. z (limits 0 → 2):

$$\int (27/4) \, dz = 27z/4 \rightarrow [27(2)/4] - [0] = 27/2$$

Answer = 27/2

Q4. $\int_0^2 \int_0^2 \int_0^2 (x^2 + y^2 + z^2) \, dx \, dy \, dz$

Step 1 — Integrate w.r.t. x (y, z constant, limits 0 → 2):

$$\int (x^2 + y^2 + z^2) \, dx = x^3/3 + xy^2 + xz^2$$

$$\text{Apply limits: } [8/3 + 2y^2 + 2z^2] - [0]$$

Step 2 — Integrate the result w.r.t. y (limits 0 → 2):

$$\int (8/3 + 2y^2 + 2z^2) \, dy = 8y/3 + 2y^3/3 + 2z^2 y$$

$$\text{Apply limits: } [16/3 + 16/3 + 4z^2] - [0] = 32/3 + 4z^2$$

Step 3 — Integrate the result w.r.t. z (limits 0 → 2):

$$\int (32/3 + 4z^2) \, dz = 32z/3 + 4z^3/3$$

Step 4 — Apply the z-limits (0 to 2):

$$= [32(2)/3 + 4(2)^3/3] - [0] = 64/3 + 32/3 = 96/3$$

Answer = 32

Method (used for every problem above): integrate innermost → apply its limits → substitute into the middle integral → apply its limits → substitute into the outer integral → apply its limits to get the final number.

5. Quick Summary

- Integration reverses differentiation and gives total accumulation, area, or the original function.
- Definite integrals have limits and give a fixed number; indefinite integrals have no limits and end in "+ C."
- Double integrals ($\iint$) integrate a function of two variables, one axis at a time.
- Triple integrals ($\iiint$) integrate a function of three variables, one axis at a time.
- Always work from the innermost integral outward, applying each variable's limits before moving to the next.